

Intel's Core family — a brief introduction

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Intel's Core family - a brief introduction

- 1. Lessons learned from the preceding Pentium 4 family
- 2. Intel's key design decisions made for the Core family
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Intel's Core family- a brief introduction

- In designing the Core family, Intel [drew on the lessons learned](#) from its earlier Pentium 4 line.

This Chapter begins by examining those lessons and the design decisions that emerged from them.

- Next, we provide an overview of the [evolution of the x86 ISA](#), which forms the foundation of the Core family.
- Finally, we present a brief [overview of the Core family](#) itself, highlighting the main processor categories it covers.

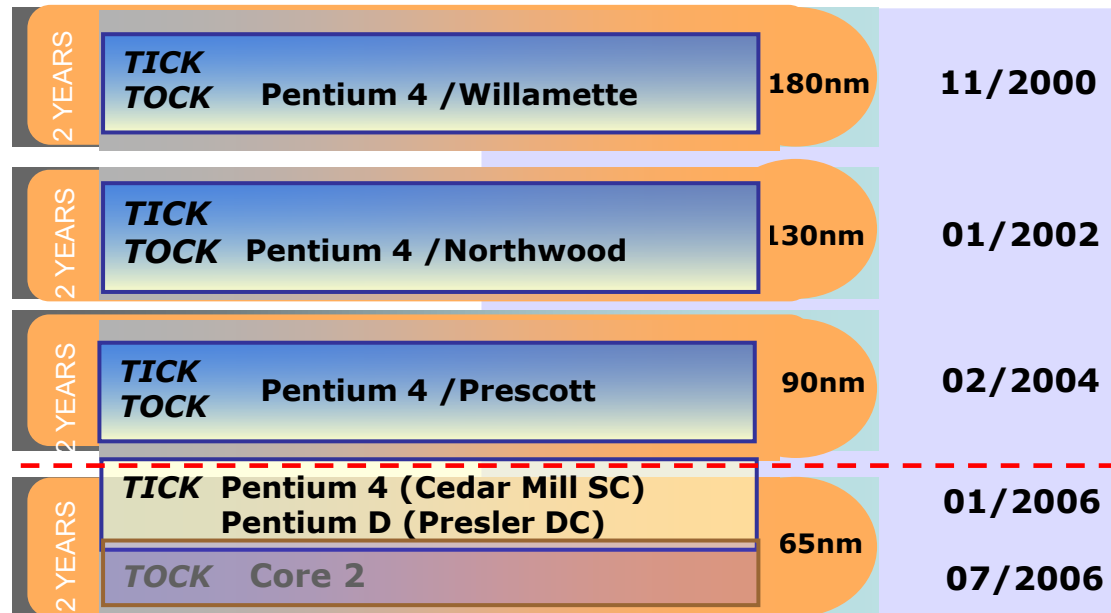
1. Lessons learned from the design of the preceding Pentium 4 family

1. Lessons learned from the design of the preceding Pentium 4 family (1)

1. Lessons learned from the design of the preceding Pentium 4 family

- The **Pentium 4 family** was launched in **11/2000**.

It had **3 generations**, as follows:



1. Lessons learned from the design of the preceding Pentium 4 family (2)

The key innovations introduced in the Pentium 4 family

Intel introduced the following **three key innovations** in its Pentium 4 family:

- a) **hyperthreading** (along with the 2. core (Northwood)) in 2002,
- b) **64-bit ISA and microarchitecture** (along with the 3. core (Prescott)) in 2004 and
- c) **dual-core processors** ((along with the 3. core (Prescott)) in 2005,

as shown below.

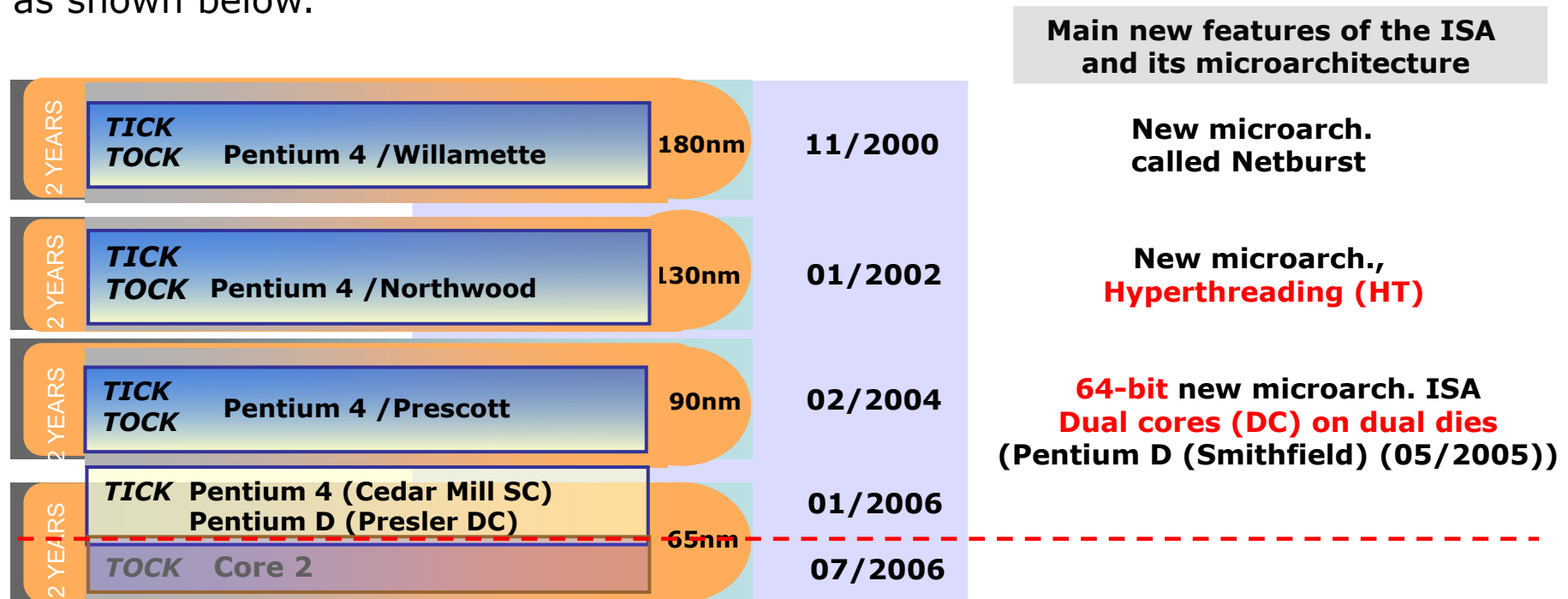


Figure: The key innovation introduced by Intel with the Pentium 4 family
The above-mentioned key innovations are detailed in the Annex (Annex 1).

1. Lessons learned from the design of the preceding Pentium 4 family (3)

The Pentium 4 debacle

- At its introduction, Intel proudly claimed that the Pentium 4 family **would reach the clock speed of 10 GHz by around 2005** [141].
- However, by **10/2004**, Intel's then-CEO (Greg Barrett) admitted that the Pentium 4 family **would not even achieve 4 GHz** [141].
- After the Pentium 4 debacle, Intel learned **two important lessons**.

1. Lessons learned from the design of the preceding Pentium 4 family (4)

Lesson one

- This shortfall was largely attributed to the **microarchitecture** of the Pentium 4 family, which prioritized **maximum clock speed** for highest performance, at the expense of excessive power consumption and heat generation.
- Intel learned from this experience and, following the Pentium 4 era, **shifted its design focus from raw performance to maximizing performance per Watt**, as indicated below.

Processor design focuses on
Highest performance
(e.g. in GFLOPS)
up to the Pentium 4 family.



Processor design focuses on
Highest performance/Watt
(e.g. in GFLOPS/Watt)
following the Pentium 4 family.

Figure: Shifting Intel's design focus while developing processors

1. Lessons learned from the design of the preceding Pentium 4 family (5)

Lesson two

- In each **Pentium 4 generation**, Intel **simultaneously** changed both the IC **technology** and the **microarchitecture**.

This type of processor development model, in which both the IC technology and the microarchitecture are changed simultaneously, is known as the **single-phase development model**.

While this approach allowed for **rapid development**, it also introduced significant **risks** – making the development process more **error-prone**.

- Recognizing these challenges, Intel transitioned with its **Core 2 processor** family to a **more structured two-phase development model**.

In this model:

- **Phase 1** focuses on introducing a **new microarchitecture** using existing IC technology.
- **Phase 2** applies **new IC technology** to the proven microarchitecture.

This staggered approach **improves reliability** and **greatly simplifies validation**.

2. Intel's key design decisions made for the Core family

2. Intel's key design decisions made for the Core family (1)

Intel's **initial design decisions** made for to the Core family.

For designing the **Core family**, Intel utilized the **lessons learned** from the Pentium 4 family, as previously discussed, such as

- a) Intel transitioned **from the performance-at any cost design approach** of the Pentium 4 family to a more balanced **performance per Watt approach**, to address the power dissipation challenges of the Pentium 4 architecture and
- b) **Intel also abandoned the** single-phase evolution model used in the Pentium 4 family and replaced it by a **two-phase Tick-Tock model, where**
 - the **Tick phase** represents a **shrink in manufacturing process technology**, (e.g. from 65 nm to 45 nm) and
 - the **Tock phase** is used to introduce a **new microarchitecture** built on the refined process.

2. Intel's key design decisions made for the Core family (2)

Intel's explanation for introducing a two-phase processor development model in the Core family

- Intel's vice president of Design Engineering Group and General Manager of SOC design (Timothy Wilson, who led the development of the Meteor Lake processor), answered this question as follows:

*"In general, **when we introduce new processes, we tend to reduce the risk in terms of architecture.** Once we achieve a stable process flow, we tend to make a bigger leap by improving the architecture."* [112]

- In other words, the introduction of the **two-phase development model aimed to reduce the complexity of designing a new processor architecture on a new IC technology.**

2. Intel's key design decisions made for the Core family (3)

Intel's initial design decisions made for to the Core family -2.

Beyond utilizing lessons learned from the Pentium 4 design, Intel made a **restriction in the implementation of the first two lines** of the Core family, as follows:

- c) **Intel skipped multithreading (called hyperthreading) in its first Core lines** (Core 2 and Penryn), likely to **speed up market entry by** simplified design.
Hyper-Threading returned later with Nehalem, once when its Core architecture became more advanced.

Subsequently, we will explore **how Intel modified its two-phase processor development model** throughout the evolution of its Core family.

2. Intel's key design decisions made for the Core family (4)

The fate of Intel's two-phase Tick-Tock processor evolution model

Intel launched the initial generations of its Core processors in alignment with its announced two-phase Tick-Tock development model, as illustrated below.

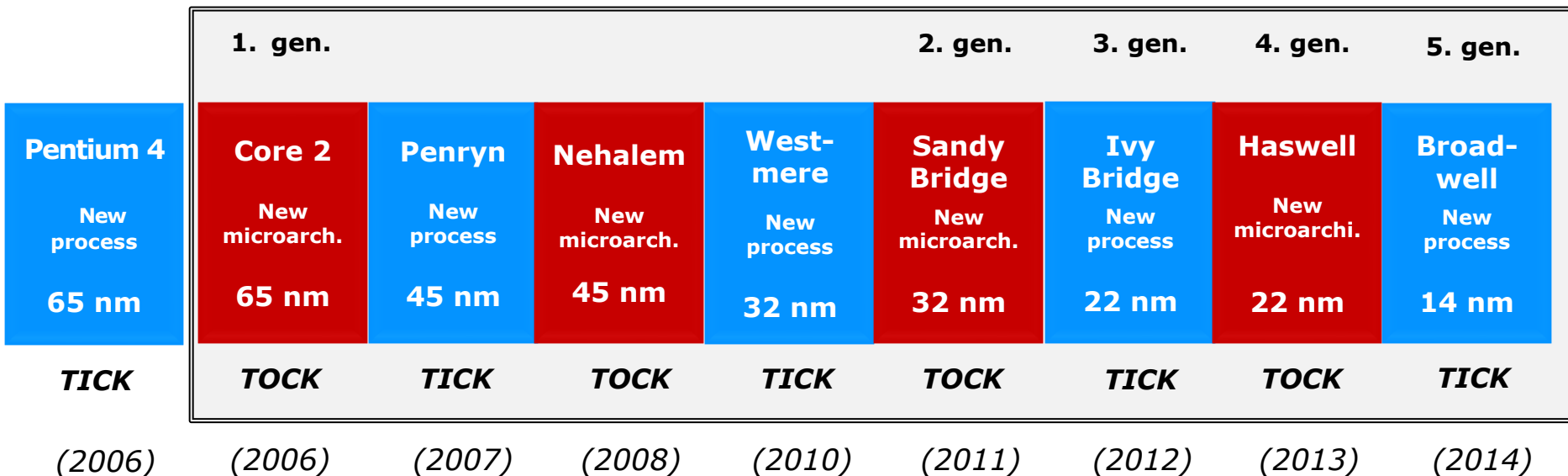


Figure: The initial generations of Intel's Core family launched in accordance to its two-phase Tick-Tock model

In the Figure above, the blue rectangles represent the Tick phases, which introduce new process technologies, while the red ones denote the Tock phases, which deliver new microarchitectures.

2. Intel's key design decisions made for the Core family (5)

Remarks on the two-year development cycle and the rate of shrinking IC technologies

- The two-year development cycle between consecutive Tick phases obviously coincides with Moore's revised law.
- Additionally, the feature size is reduced by a rate of 0.7 since at this rate
 - the same number of transistors can be implemented on a die size that is roughly halved or
 - the transistor count can be roughly doubled on the same die size.

2. Intel's key design decisions made for the Core family (6)

Remarks on the irregularity in naming Core family generations

- The **Core 2** line marked the beginning of Intel's Core family, and became its **1st generation**.

Surprisingly, the **Sandy Bridge** line was later designated as the **2nd generation**, after which Intel adopted a **sequential numbering system**.

- Notably, the **Nehalem and Westmere lines** – released between Core 2 and Sandy Bridge – **were not assigned generation identifiers**, creating a curious gap in Intel's naming scheme.

This irregularity raises some questions about Intel's naming conventions.

- A likely explanation lies in the **origins of the designs**: both Core 2 and **Sandy Bridge** were developed at Intel's **Haifa Design Center**, while **Nehalem and Westmere** came from the **Oregon Design Center**.

It seems that when Intel labeled the Sandy Bridge as the 2nd generation, it chose **to omit the intervening designs from the generational count**.

- It is worth noting that **in 2023, Intel abandoned its traditional generation-based naming scheme with the launch of its Meteor Lake line**, introduced the **Core Ultra Series x** designation for its laptop processors.

2. Intel's key design decisions made for the Core family (7)

Intel's unexpected stumbling in mastering the 10 nm technology

- Back in 2011, in its technology roadmap, Intel announced its plan to introduce the 10 nm technology by 2015, as shown below.

Intel Technology Roadmap

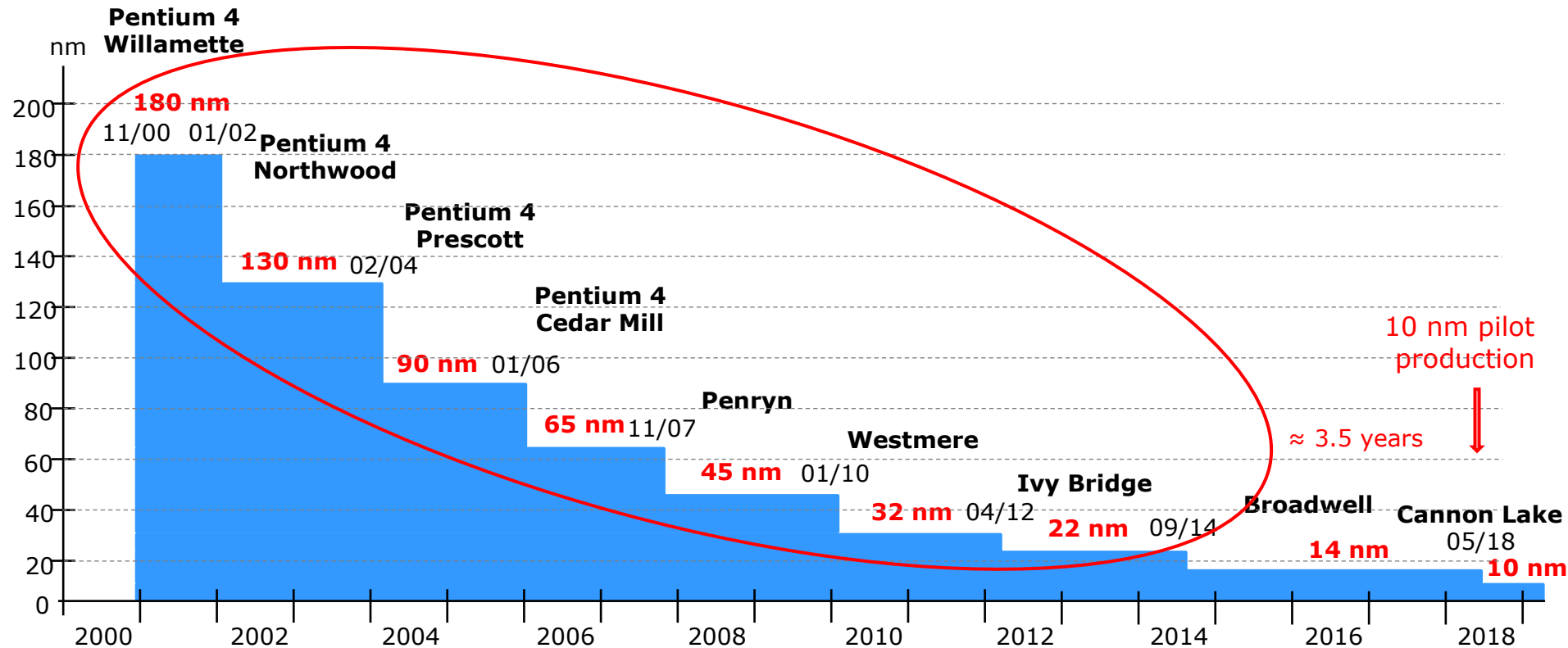
Process Name	<u>P1266</u>	<u>P1268</u>	<u>P1270</u>	<u>P1272</u>	<u>P1274</u>
Lithography	45 nm	32 nm	22 nm	14 nm	10 nm
1 st Production	2007	2009	2011	2013	2015

Table: Intel's IC technology roadmap from 2011 [142]

- However, the company faced unexpected challenges in perfecting the 10 nm process, which led to a significant delay in meeting its goal.
 - It wasn't until 2018 that Intel announced the availability of its 10 nm process as a pilot implementation, to be discussed later.
- Full-scale mass production began one year later, in 2019.

2. Intel's key design decisions made for the Core family (8)

Intel's IC technology transitions from 2000 to 2018



- The Figure above illustrates that Intel consistently maintained an approximately 2-year development cycle, while shrinking feature sizes by a factor of about 0.7 up to its Ivy Bridge line.
- However, as previously noted, Intel encountered a significant delay in mastering the 10 nm technology.

2. Intel's key design decisions made for the Core family (9)

Moving away from the two-phase development model and including additional “optimization phases” after the Tock phase [27]

- To bridge the delay in transitioning to the 10 nm process, Intel abandoned its traditional 2-year development model following the 6th Core generation.
- In its place, the company adopted a revised model that inserts one or more optimization phases after the Tock phase, as illustrated below.

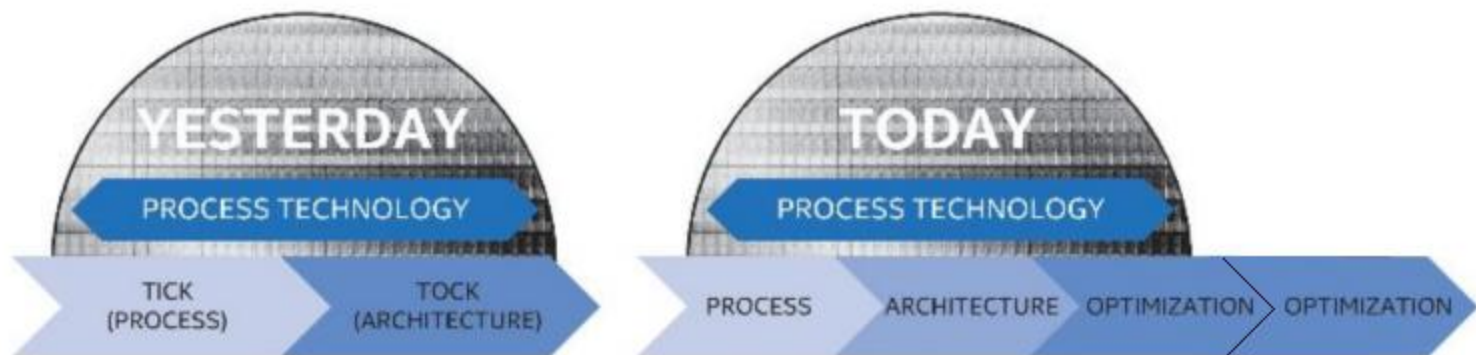


Figure: Intel's switch to a revised development model that inserts additional optimization phases after the Tock phase [28]

- Accordingly, following the 6th generation Tock line, known as the Skylake line, Intel inserted several optimization phases into the Core family's evolution path, as depicted in the the next Figure.

2. Intel's key design decisions made for the Core family (10)

Optimization phases inserted following the 6th generation of the Core family

1. gen.		2. gen.		3. gen.	4. gen.	5. gen.	
Core 2 New microarch. 65 nm	Penryn New process 45 nm	Nehalem New microarch. 45 nm	Westmere New process 32 nm	Sandy Bridge New microarch. 32 nm	Ivy Bridge New process 22 nm	Haswell New microarchi. 22 nm	Broadwell New process 14 nm
<i>TICK</i>		<i>TOCK</i>	<i>TICK</i>	<i>TOCK</i>	<i>TICK</i>	<i>TOCK</i>	<i>TICK</i>
(2006)	(2007)	(2008)	(2010)	(2011)	(2012)	(2013)	(2014)

6. gen.	7. gen.	8. gen. ¹	9. gen.	10. gen.
Skylake New microarch. 14 nm	Kaby Lake New microarch. 14 nm	Kaby Lake R/G Coffee Lake Amber Lake-Y Whiskey Lake-U Cannon Lake (10 nm) 14 nm	Coffee Lake R New microarch. 14 nm	Comet Lake New microarch. 14 nm
TOCK	Optimization	Optimization	Optimization	Optimization
(2015)	(2016)	(2017/18)	(2018)	(2019)

¹The 8th generation encompasses the following processor lines:

- Kaby Lake Refresh
- Kaby Lake G with AMD Vega graphics
- Coffee Lake
- Amber Lake Y
- Whiskey Lake U
- (all 14 nm) and
- **Cannon Lake (10 nm)**

lines [29].

R: Refresh

2. Intel's key design decisions made for the Core family (11)

Optimization phases inserted following the 6th generation of the Core family -2

- The **optimization phases** inserted were built on the **14 nm technology**, with the sole **exception** being the **10 nm Cannon Lake line**.
- Released in **05/2019**, the **Cannon Lake line** served primarily as a **pilot implementation to showcase Intel's 10 nm process capability**.

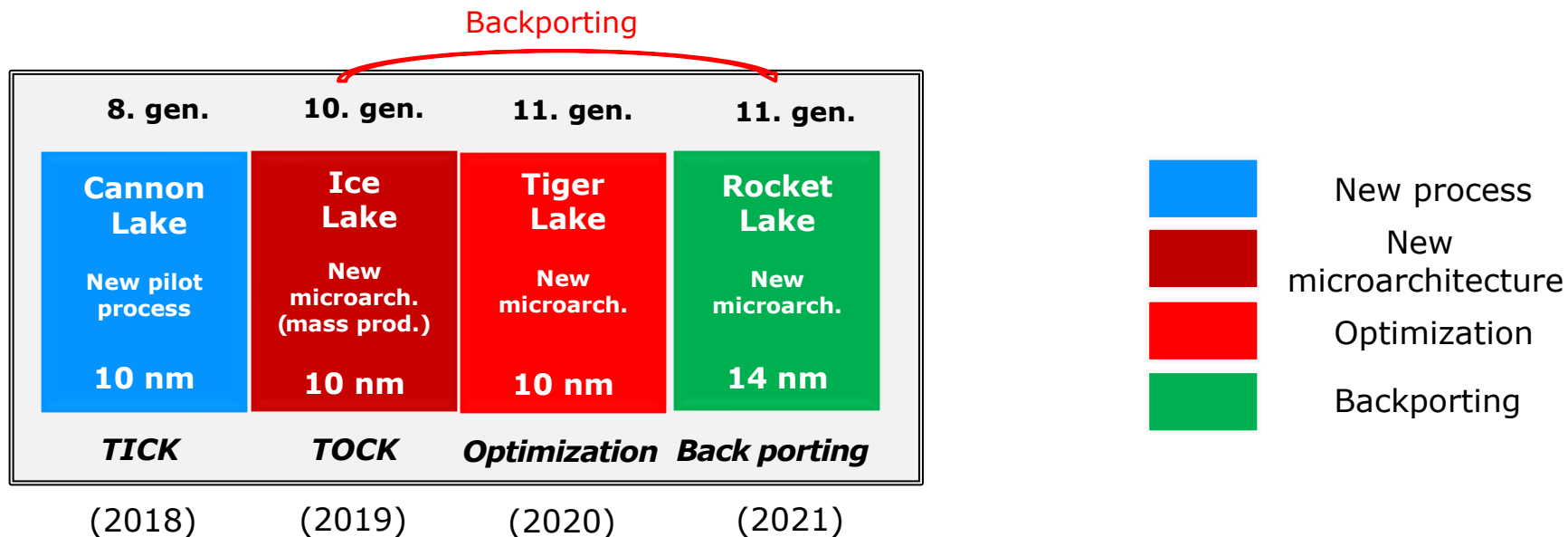
It included only a **single, modest performance model** with just dual cores and didn't include a GPU.

- For the **mass production** of 10 nm processors, Intel required an **additional year**.

2. Intel's key design decisions made for the Core family (12)

Optimization phase and backporting following the 10th generation

Symmetrical multicores



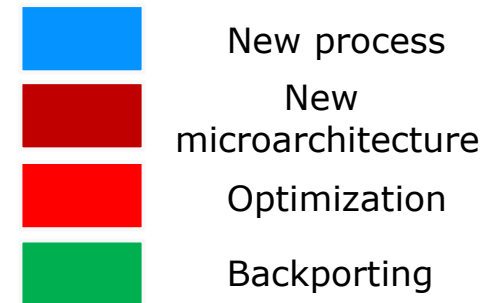
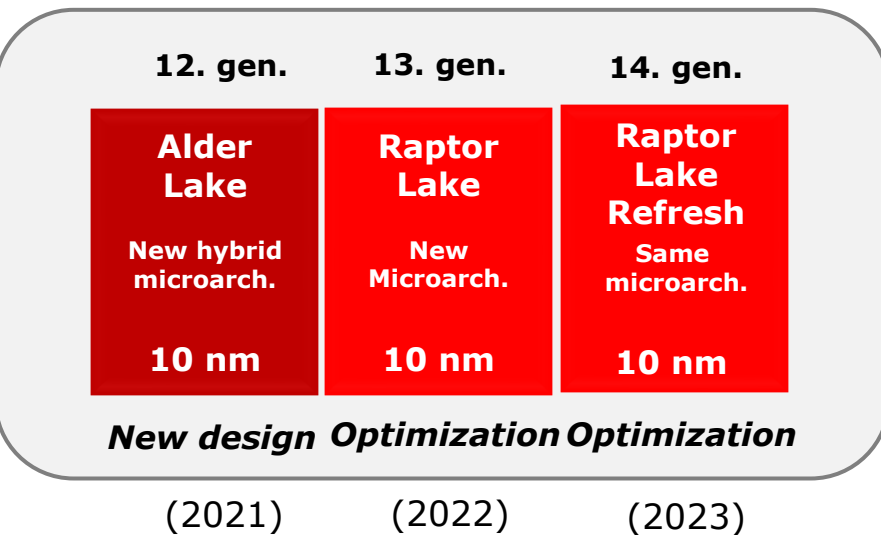
- As shown above, Intel's 11th generation of the Core family belongs to the **optimization phases**, whereas the 11th gen. **Rocket Lake line** implements **backporting**.
- Backporting** refers to the **implementation** of an advanced microarchitecture **on an older, less advanced technology**, typically to reduce manufacturing costs.
- We point out that all Core processors up to the 11th generation are **symmetrical multicores**, meaning that **each core has the same capabilities**, allowing straightforward core scheduling and power management.

2. Intel's key design decisions made for the Core family (13)

Introducing big.little designs beginning with the 12th generation of the Core family

- Along with its 12th gen. Alder Lake line Intel switched from “**symmetrical multicores**” to “**big.little**” architectures, to reduce power consumption.

(Single-die) big. little designs



- Power consumption is reduced by including lower power-consuming, less powerful (little) cores as well, which the OS schedules for less demanding tasks, as discussed in the Section “Intel’s Core-based client processors.
- The 12th-generation Alder Lake line was succeeded by two optimization phases (known as generations 13 and 14), which are still manufactured using the 10 nm process.

2. Intel's key design decisions made for the Core family (14)

Introducing heterogeneous multi-die big.little designs big.little designs after the 14th generation of the Core family -1

- Following its 14th gen. Raptor Lake Refresh line, Intel made a significant shift in its Core processor architecture.

The new design moves away from the traditional monolithic implementation and adopts a heterogeneous multi-die big.little design to reduce manufacturing costs.

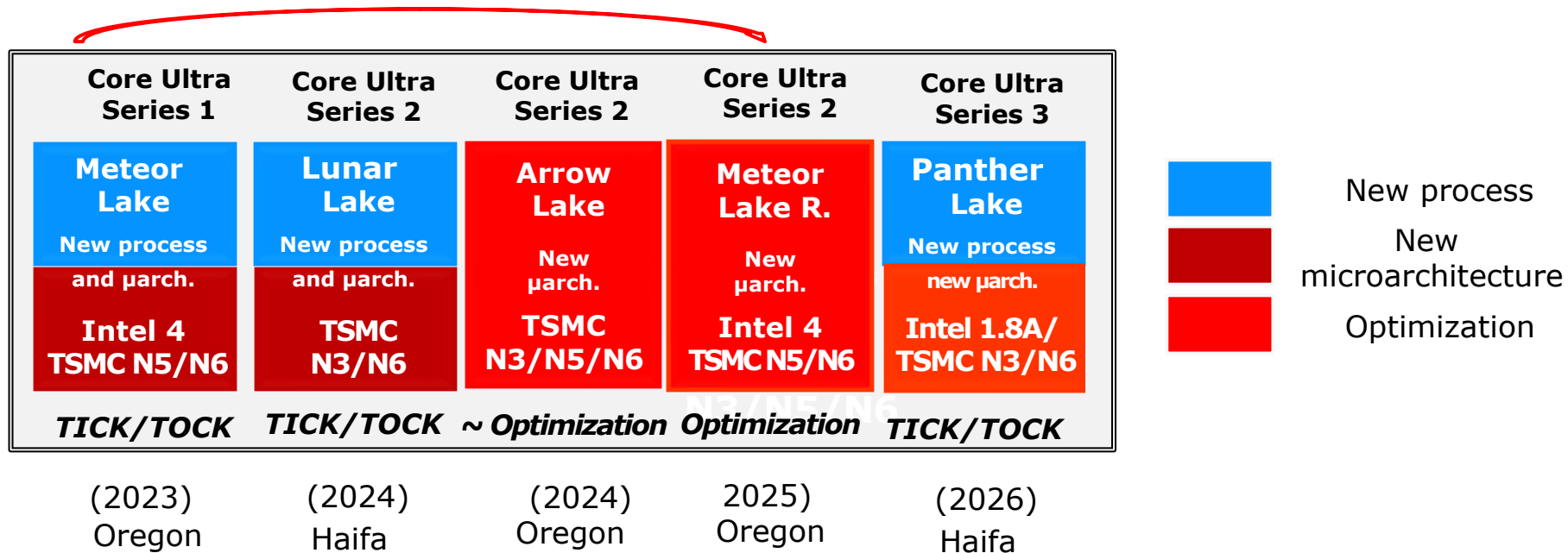
- A heterogeneous multi-die processor is built up of multiple dies with different functionalities, such as Compute die or Graphics die.

By fabricating less demanding dies, such as IO dies, using less advanced and thus less costly processes, the overall fabrication costs become reduced.

2. Intel's key design decisions made for the Core family (15)

Introducing heterogeneous multi-die big.little designs big.little designs after the 14th generation of the Core family -2

Multi-die heterogeneous big. little designs



- To emphasize the innovative nature of the new design, Intel has rebranded its client lines to "Core Ultra" processor lines.
- Additionally, the company has replaced the term "generation" with the "Series" designation, starting with Series 1, marking its first heterogeneous big.little line, referred to as the Meteor Lake line.

2. Intel's key design decisions made for the Core family (16)

Single-phase heterogeneous multi-die designs

- Even after mastering the technical challenges of the 10 nm process Intel was unable to recapture its previous momentum in advancing IC technology.

Instead, the company's process and microarchitecture development followed an irregular trajectory, at times progressing both technology and microarchitecture simultaneously in a single phase, to reduce time to market, as shown in the previous Figure.

2. Intel's key design decisions made for the Core family (17)

Single-phase heterogeneous multi-die designs

- The previous Figure shows that Meteor Lake's, Lunar Lake's, and Panther Lake's deviated from Intel's traditional two-phase development model and shifted to single-phase designs.

In single-phase designs, both the IC technology and the microarchitecture are changed simultaneously.

- This approach increases development risks, but reduces the time required to bring the product to market.
- It is likely that the company felt pressured to shorten the time to market due to the decline in its income during this time.

2. Intel's key design decisions made for the Core family (18)

Remark on using TSMC's technology to manufacture Intel processors

- Originally, Intel intended to fabricate the CPU dies of its **Lunar Lake and Arrow Lake processor lines** using its own **20A** and **18A nodes**.

However, the company encountered **unexpected challenges** in developing these advanced processes.

To address its manufacturing setbacks, **Intel cancelled developing the 20A process** and redirected its effort toward the 18A node.

- At the same time, Intel **asked TSMC to produce its Lunar Lake and Arrow Lake processor dies** to avoid delays in market entry.

Specifically, the compute dies of both lines were fabricated using TSMC's most advanced available process, the N3 process.

- Finally, the CPU dies of Intel's Core Ultra Series 3 processors could be fabricated using the company's own 18A process.

3. Evolution of the x86 ISA

3. Evolution of the x86 ISA (1)

3. Evolution of the x86 ISA

- Since its inception in 1978, the x86 ISA has undergone significant evolution in many aspects, for instance, to increase computing efficiency through the introduction of SIMD (vector) instructions, to enhance security by providing memory protection etc.
- From those mentioned, in this Section we will focus solely on the SIMD extensions.
- SIMD extensions, also known as vector extensions, enable the simultaneous execution of the same operation on multiple pairs of data.

This parallelism significantly speeds up computations, as illustrated below for a 4-wide SIMD operation.

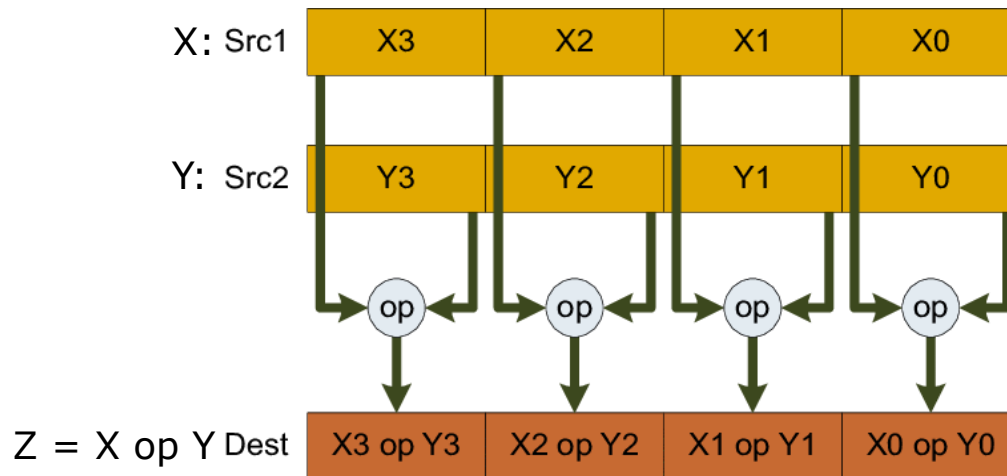


Figure: Interpretation of 4-wide SIMD operations [33]

SIMD: Single Instruction Multiple Data

3. Evolution of the x86 ISA (2)

Historical remarks to the emergence of SIMD processing in Intel's processor lines -1

- In the mid-1990s, the rise of multimedia applications created a growing demand for 2D graphics support.

In 2D graphics, objects are represented by a large number of pixels, each defined by its three RGB color components and a transparency value, typically encoded as FX8 data.

Consequently, in multimedia each pixel is represented by FX32 data, therefore to perform efficient computations in 2D graphics architectural support for FX-SIMD operations became essential.

- In 1997, Intel responded to the growing demand for multimedia processing by introducing the MMX (Multi-Media Extension) ISA enhancement in its 32-bit Pentium-MMX processor.

MMX extended the ISA with 64-bit FX SIMD capability, enabling the processor to execute two FX32 operations per cycle.

However, MMX did not introduce dedicated SIMD registers; instead, it utilized the 64-bit mantissa portion of the existing 8 FP registers.

This solution however, prevented simultaneous SIMD and FP processing, limiting parallelism in mixed workloads.

Transparency (átláthatóság): The extent to which a pixel allows light or background imagery to pass through it, ranging from fully opaque to fully transparent

3. Evolution of the x86 ISA (3)

Historical remarks to the emergence of SIMD processing in Intel's processor lines -2

- In the late 1990s, 3D graphics emerged as a significant advancement in computer visualization.
Unlike 2D graphics, 3D objects are represented by triangles, each defined by three vertices.
These vertices are encoded as FP32 data with additional attributes, such as its color and transparency.
- Processing 3D graphics requires extensive FP32 computations, which can be significantly speeded up using FP SIMD operations.
- To support 3D graphics processing, Intel introduced the SSE (Streaming SIMD Extensions) in 1999, enabling 128-bit FP SIMD processing within 8x128-bit dedicated SIMD registers.
Each 128-bit FP SIMD instruction could operate on four FP32 operand pairs.
Intel first implemented the SSE extension in its Pentium III processor in 1999.

3. Evolution of the x86 ISA (4)

Historical remarks on the emergence of integrated graphics in Intel's processors

To support the growing demand to run multimedia and 3D graphics workloads Intel

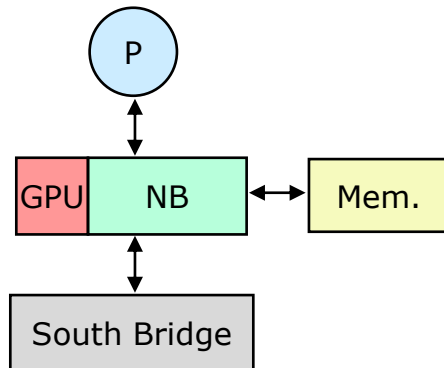
integrated graphics into its processors in **three major steps**:

first, it **integrated GPUs into the NB**, next, Intel **integrated the GPU and CPU dies within the processor package**; finally, **in 2011**, Intel **integrated the GPU onto the CPU die**, as shown below.

Implementing integrated graphics in Intel's processors

In the North Bridge (NB) integrated GPU

The NB is also called GMCH (Graphics and Memory Control Hub)

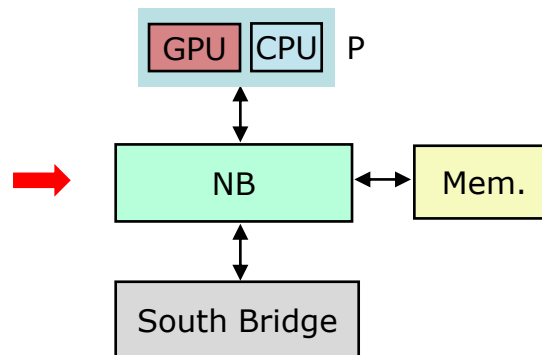


Intel's Pentium III to Nehalem processors (1999 – 2009), first in Intel's 810 NB (1999) then in subsequent implementations

Graphics generations 1-4

On-package integrated GPU (MCP)

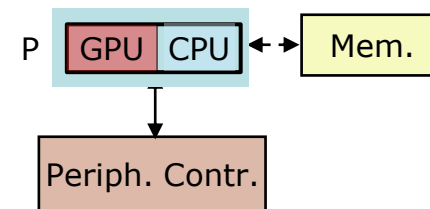
CPU and GPU are on separate dies, cased in a multi-chip package (MCP)



Intel's Westmere-based client processors (1/2010)

Graphics generation 5

On-die integrated GPU



Intel's Atom-based Pineview (LT, 1/2011) and subsequent processors
Intel's Sandy Bridge (DT, 1/2011) and subsequent client processors

Graphics generations 6 and above

LT: Laptop
DT: Desktop

3. Evolution of the x86 ISA (5)

Enhancing the register space available for the x86 SIMD ISA extensions [34] -1

- The **SSE** extension had an **8x128-bit register space**, as discussed so far, it was **extended** in the following **three steps**:
 - first to **16x128-bit** (in the **SSE3** extension) (in the 64-bit mode), in **2004**,
 - then to **16x256-bit** (in the **AVX** extension, in **2011**) and
 - finally, to **32x512-bit** (in the **AVX512** extension, in **2017**),.

as illustrated below.

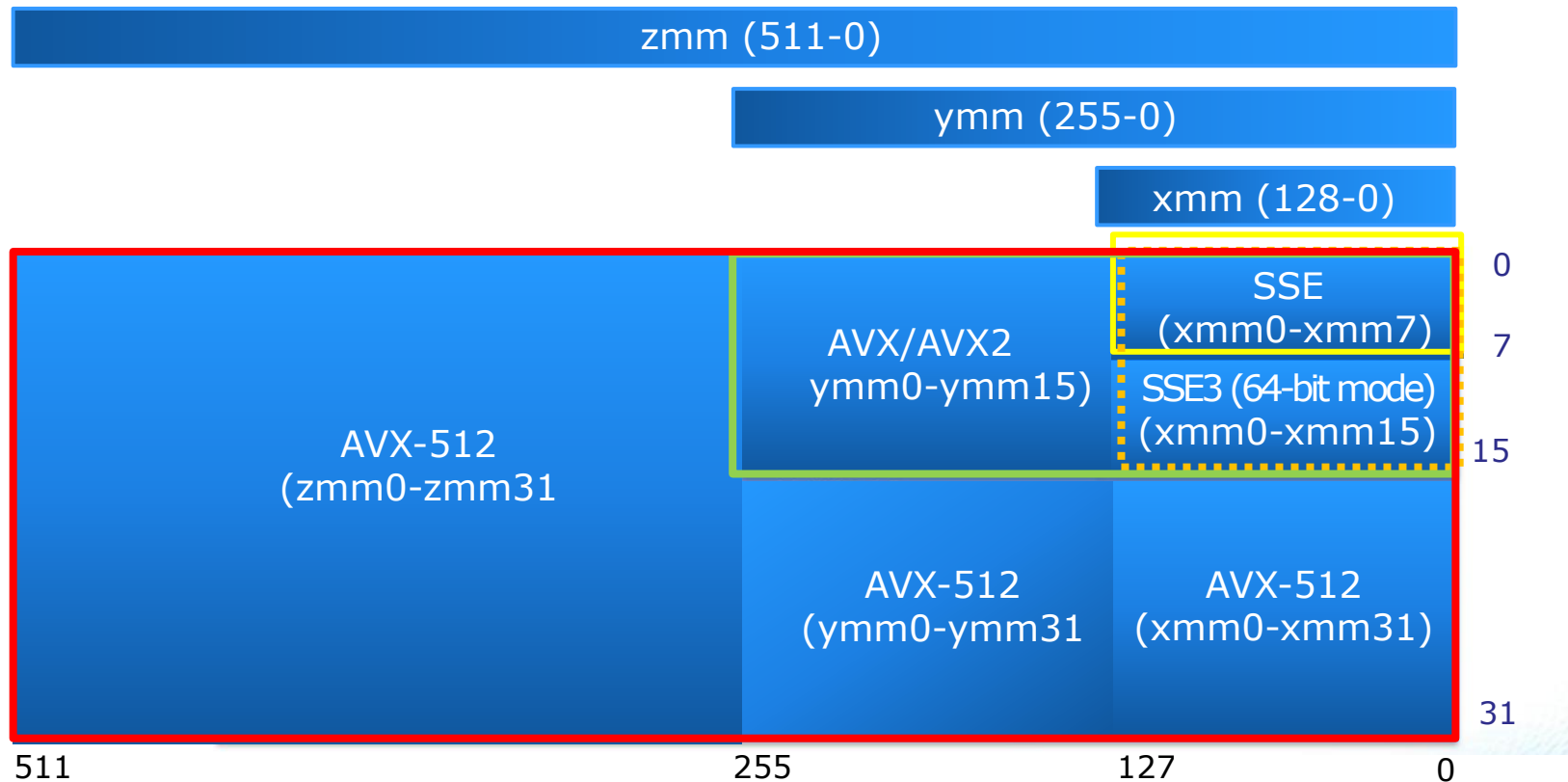


Figure: Dedicated register spaces provided for the x86 ISA SIMD extensions [35]

3. Evolution of the x86 ISA (6)

The peculiarity of expanding the x86 register space for SIMD processing [34] -

- Subsequent **register extensions in the x86 ISA** were implemented using **aliasing**.
- This means that the preceding **half-long registers are used as the lower portions of the extended registers**, e.g., the XMMx registers, are used as the lower parts of the YMMx registers, as illustrated below.

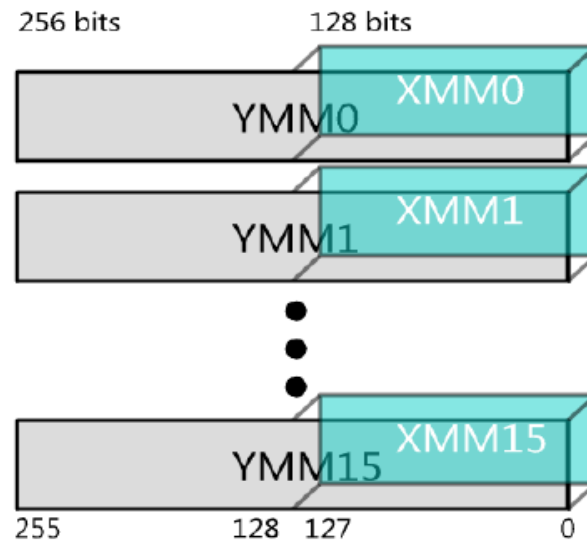


Figure: Example for aliased extension: the 128-bit XMM registers are used as the lower 128 bits of the 256-bit YMM registers. [36]

Aliasing: Azonosítás (azonos adatokra való hivatkozás két eltérő névvel)

3. Evolution of the x86 ISA (7)

Summing up Intel's x86 SIMD extensions -1

- Along with the introduced dedicated register spaces, Intel introduced one after another the following **SIMD extensions**:

64-bit FX	(MMX)	to support multimedia (MM) processing
128-bit FP	(SSE)	to support 3D processing
128-bit FX/FP	(SSE2-4)	to support both MM and 3D processing
256-bit FP	(AVX)	primarily to give better support for 3D processing
256-bit FX/FP and	(AVX2)	to enhance support both for MM and 3D processing
512-bit FX/FP	(AVX512)	to provide more enhanced support for MM and 3D processing

- One important aspect of Intel's SIMD extensions is that **each new extension, such as MMX, SSE, AVX, AVX2 or AVX512 comes with its own distinct set of instructions.**

Therefore, if application aim to take advantage of the increased width and computational efficiency of longer SIMD extension supported in processors, must be explicitly rewritten and recompiled.

- For a comprehensive overview of Intel's x86 extensions, see the following Table and Figure.

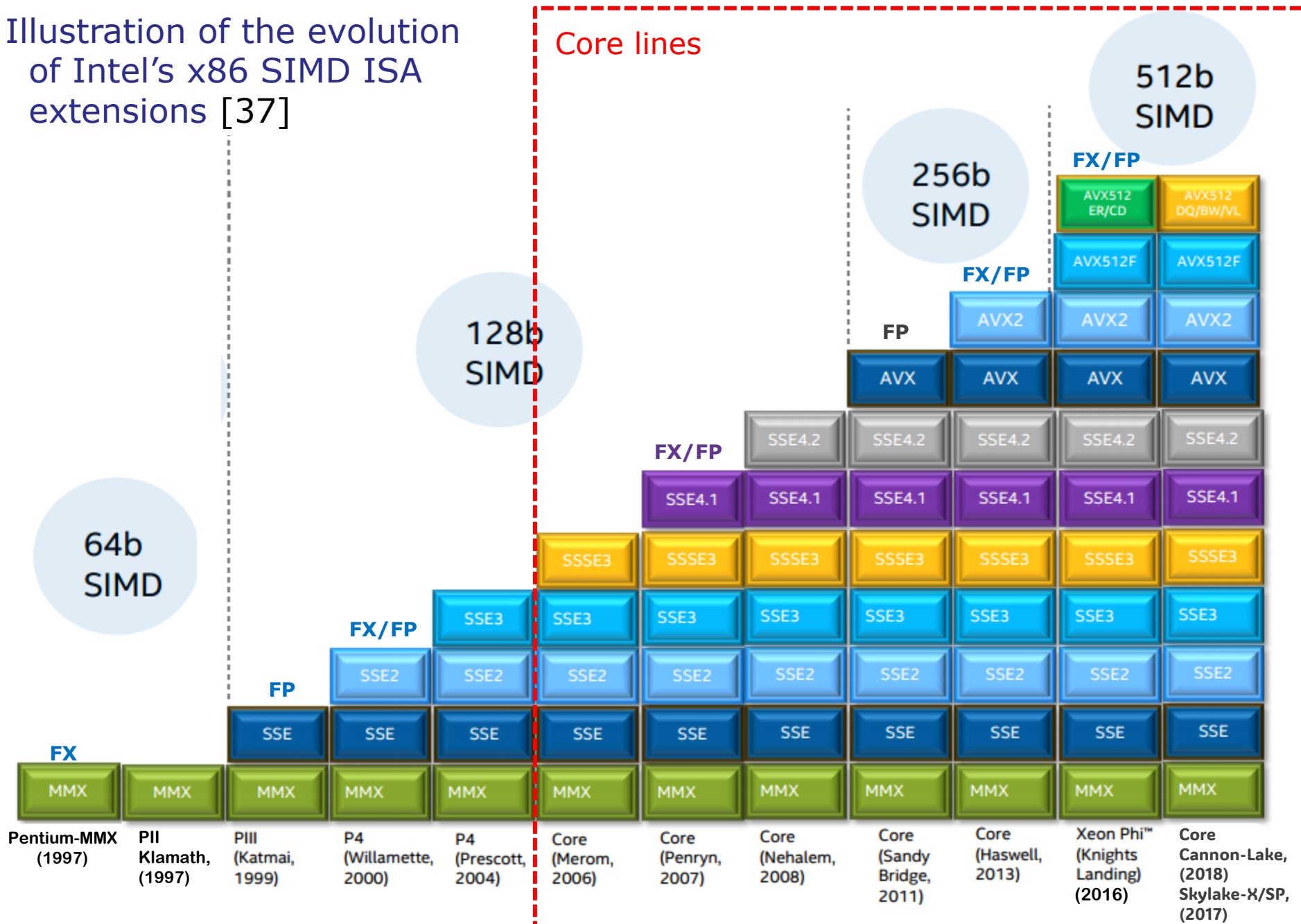
3. Evolution of the x86 ISA (8)

Timeline and first implementation of Intel's x86 SIMD extensions

Introduction and the long way of the evolution of SIMD support in Intel's processors					
SIMD	Extension	Processor	Width	Year	Register space/Remarks
64-bit FX	MMX	Pentium-MMX	32-bit	1997	8x64-bit MMX registers, aligned with the mantissa part of the FP registers
128-bit FP	SSE	Pentium III	32-bit	1999	8x128-bit Xmm registers
128-bit FX/FP	SSE2	Pentium 4 Willamette	32-bit	2000	
	SSE3	Pentium 4 Prescott	32-bit	2004	
	SSE3	Pentium 4 Prescott F-series	64-bit	2004	
	SSE4	Core Merom (1 st gen.)	64-bit	2006	16x128-bit Xmm registers
256-bit FP	AVX	Core Sandy Bridge (2 nd gen.)	64-bit	2011	16x256-bit Ymm registers
256-bit FX/FP	AVX2	Core Haswell (4 th gen.)	64-bit	2013	16x256-bit Ymm registers
512-bit FX/FP	AVX-512	Core Cannon Lake (8 th gen.)	64-bit	2018	32x256-bit Zmm registers

3. Evolution of the x86 ISA (9)

Illustration of the evolution of Intel's x86 SIMD ISA extensions [37]



4. Overview of the Core family

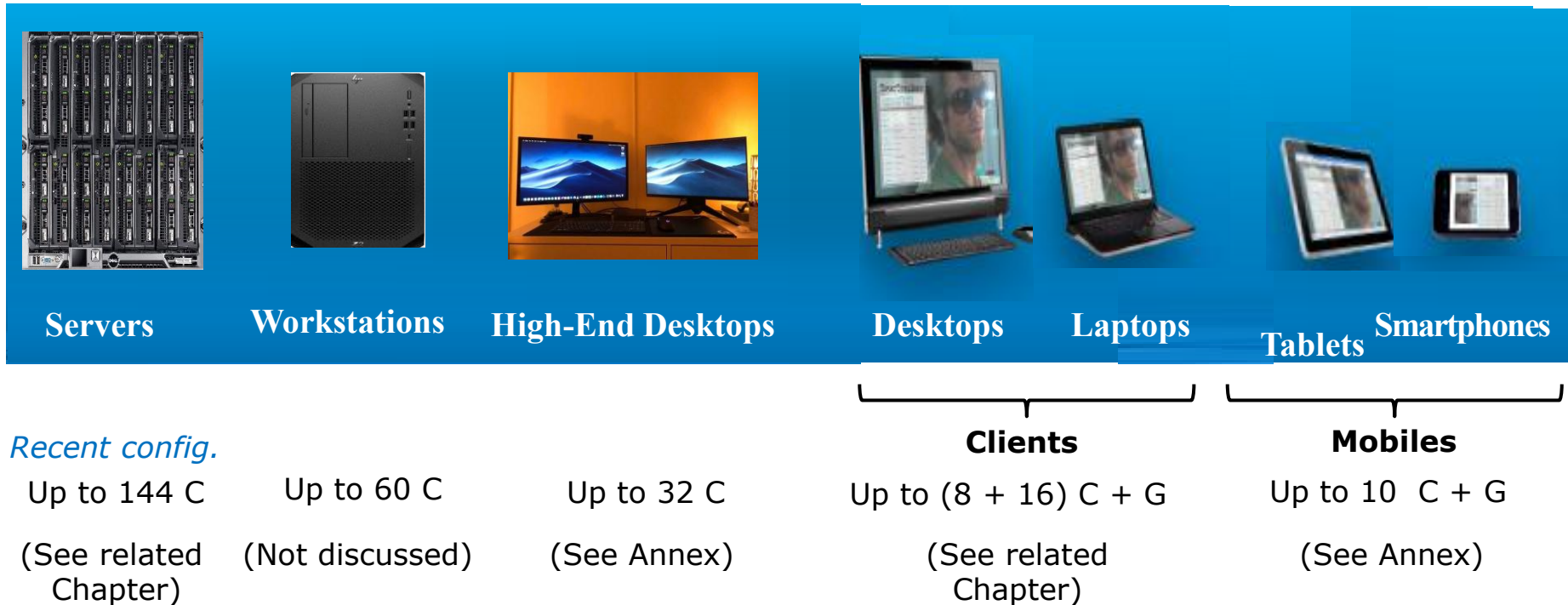
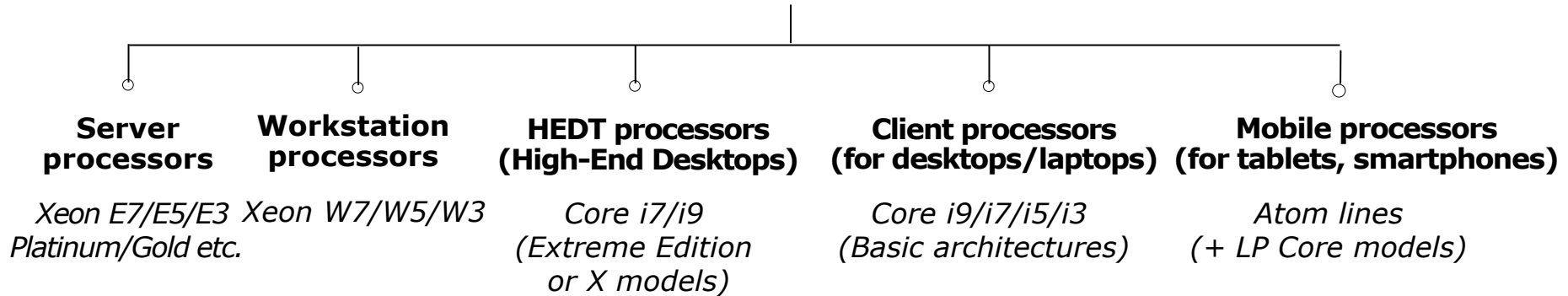
4. Overview of the Core family

The **Core Family** covers **several processor categories**, as shown in the following Figure.

4. Overview of the Core family (2)

The main processor categories in Intel's Core family

Main processor categories in Intel's Core family



Note that both Intel and AMD designate **laptop processors as mobile processors**.

4. Overview of the Core family (3)

Covering the main processor categories of Intel's Core family in the Lecture

- Intel's **Core-based server and client processor lines** are discussed in detail **in two chapters** of this lecture.
- A brief overview of Intel's **HEDT (High-End Desktop) and mobile processor** lines is provided in the **Annex**.
- The **workstation processor category** is **not covered separately**, as these processors largely share architectural and functional characteristics with Intel's server-class CPUs.

4. Overview of the Core family (4)

Remark: Interpretation of the term TDP (Thermal Design Power)

- The TDP (Thermal Design Power) of a processor (package) can be interpreted as the maximum power consumed by the processor when running realistic, power-intensive applications.

It is typically specified in the processor's datasheets and is usually given in W. It's important to note that realistic, power-intensive applications should not be confused with power viruses, which are written to draw maximum power.

- The importance of TDP lies in the fact that the power budget allocated to a processor directly limits its performance.
- Another interpretation of TDP is that it serves as a reference value for designing the platform's cooling system.
- The cooling system, also known as the thermal solution, is crucial for managing the temperature of the transistors on the die.
- It ensures that their junction temperatures (T_j), do not exceed a specified limit (T_{jmax}), which is for instance, 90°C, even while the processor dissipates up to TDP Watts.

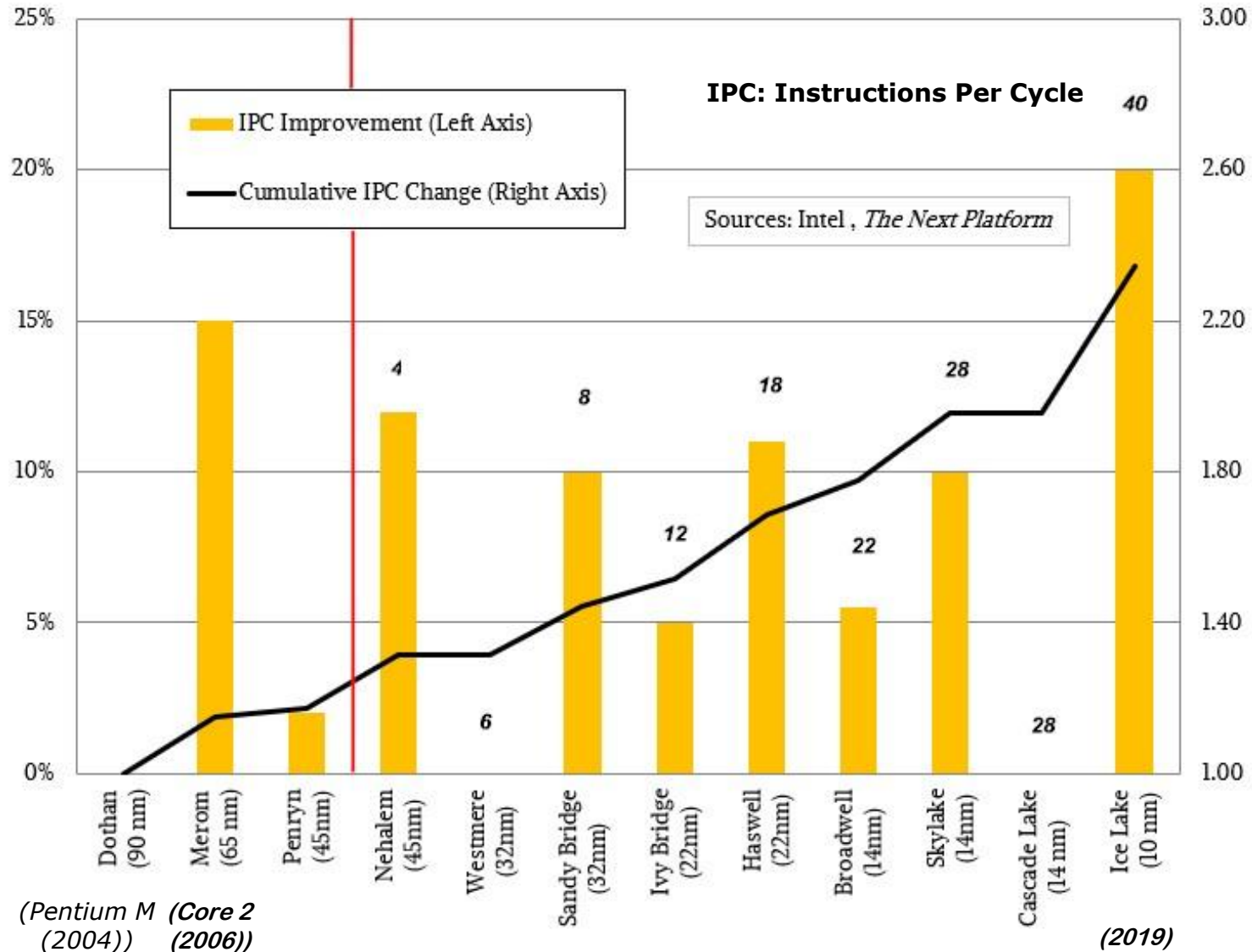
4. Overview of the Core family (5)

Typical TDP values in different processor categories of the Core line (as of now)

Processor category		TDP (as of now)	Intel's usual tags
Servers		≈ 200-400 W	
HEDTs		≈100-150 W	X
Desktops	High perf.	≈70-125 W	K
	Mainstream	≈50-65 W	S
	Low power	≈35-45W	T
Notebooks	High perf.	≈45 W	H/HQ
	Mainstream	≈25-35 W	U
	Ultra-thin	≈15 W	U
Tablets		≈5 W	Y/m

4. Overview of the Core family (6)

Intel's progress in increasing IPC in their Core family up to 2019 [32] -1



The Figure indicates a more than twofold increase in IPC within Intel's Core line in about 15 years.

4. Overview of the Core family (7)

Intel's progress in increasing IPC in their Core family up to 2019 [32] -2

Concerning the Figure, please note that Intel selected the early, mobile-oriented [Pentium M](#) processor (codenamed Dothan, 2004) as its [reference point](#).

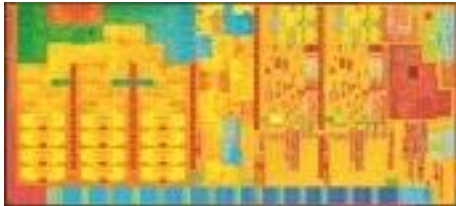
This choice may lead to a [slightly misleading and overly optimistic](#) interpretation of the results.

4. Overview of the Core family (8)

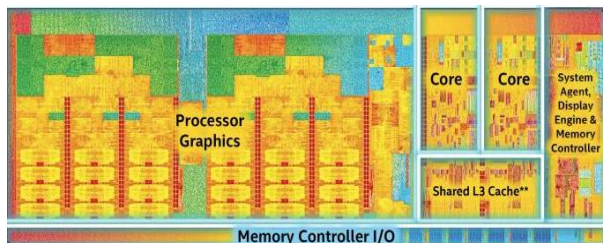
Comparing die sizes of Intel's 14 nm client and server processors - Examples (relative on scale) [38], [39]

Client dies

5th gen. Broadwell (2C, 82 mm²)
(with 5500 GPU)

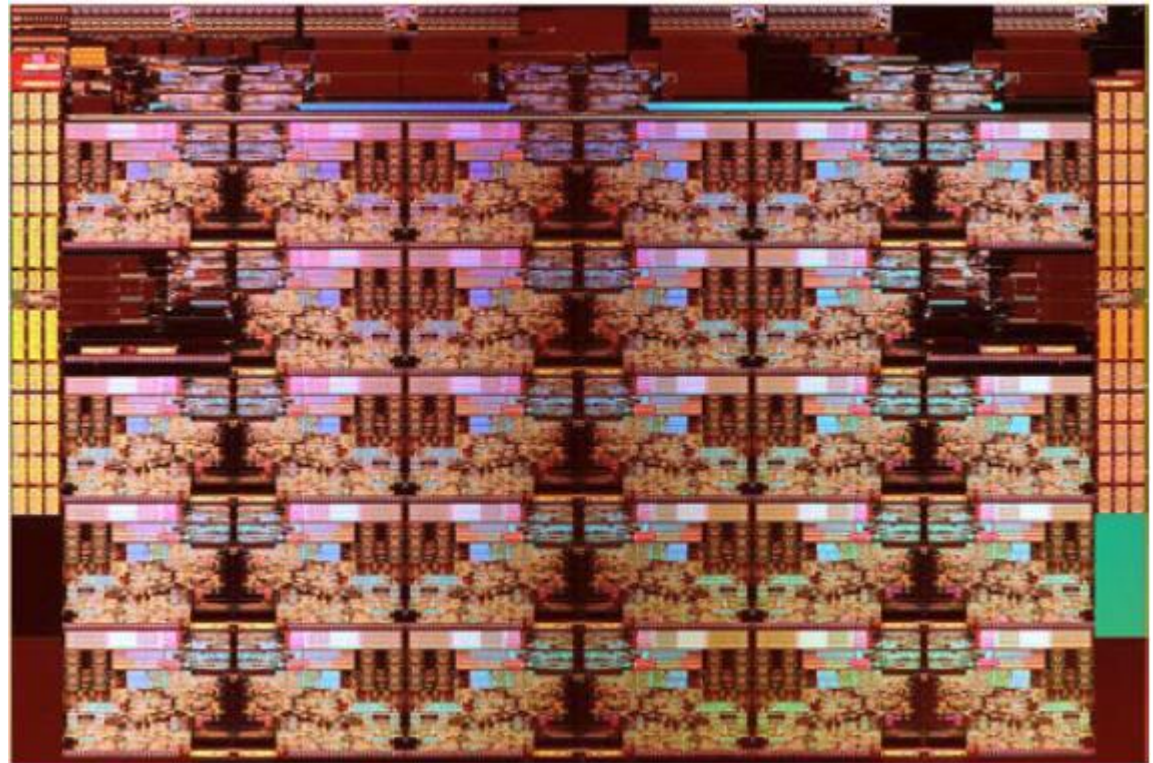


5th gen. Broadwell (2C, 133 mm²)
(with 6100/6200 GPU)



Server die

6th gen. Skylake-SP (28C, ~680 mm²)



As shown above, server processor dies are considerably larger than those of client processors.

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